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Paleomagnetic investigations of Jurassic and Lower Cretaceous sedimentary rocks of Siberia

G. A. Pospelova, G. Ya. Larionova and A.V. Anuchin

ABSTRACT: The paper contains recent information on the history of the earth's magnetic field in the Jurassic and Lower Cretaceous, obtained in paleomagnetic exploration of the key paleontological sections of sea deposits in the Bay of Anabar and Khatanga basin. Six paleomagnetic zones in the Valanginian were found for the first time, as well as alternation of paleomagnetic zones in Middle and Lower Jurassic rocks. Comparison of paleomagnetic sections of different regions indicates inversions of the magnetic field of the earth. The pole position was calculated for the Lower Valanginian: λ = = 174° E; φ = 63.2° N. Possibility is seen for the use of rocks possessing high magnetic viscosity and low values of Q and of the collapse field, for paleomagnetic purposes. -- Authors.

* * *

Paleomagnetic investigations of the Jurassic and Lower Cretaceous sedimentary rocks of certain areas of Siberia have been made for the purpose of studying the history of the geomagnetic field, preparing a paleomagnetic reference section through the Jurassic and Lower Cretaceous series, examining the possibilities for correlating sedimentary sequences of this age and ascertaining the paleolatitudes of the regions investigated.

Analysis of the data from all over the world shows that the Jurassic and Cretaceous periods are the least well known, from the paleomagnetic standpoint. The latest survey of the results of paleomagnetic studies (7) cites ten reliable determinations of the pole for the Jurassic and six for the Cretaceous periods. At present, there are several more determinations relating to the Cretaceous and Jurassic periods (1-5, 9, 11, 12, 14). Only four publications in all are devoted to the paleomagnetism of the Early Cretaceous, and Late and Middle Jurassic epochs. Yet even these few data are contradictory and hard to compare, since the rocks investigated were in different continents and their ages were not always dated with sufficient precision. The world map sketches in Figure 1 show the locations of the geomagnetic poles in the Cretaceous and Jurassic, according to the results of studies by different authors. The wide scattering of these points has several possible explanations: the non-dipole character of the geomagnetic field, continental drift, inaccurate dating of the samples and unreliability of the results.

The study of Mesozoic rocks, apart from the history of the earth's magnetic field, is of great interest also from the standpoint of the continental-drift hypothesis, since, according to the "mobilists," the present positions of the continents were determined in the middle of the Mesozoic era. Determination of paleolatitudes by paleomagnetic methods also yields information of importance to paleoclimatology and paleogeography. For these reasons, it is quite natural that the paleomagnetic investigation of the Jurassic and Lower Cretaceous rocks of Siberia is currently a very urgent matter and a subject of intense interest, the more so because our investigations are based on marine sedimentary deposits in paleontologically well characterized stratigraphic key sections.

The samples for our paleomagnetic investigations were selected in three regions of Siberia:

1) in the Khatanga basin, on the Boyarka river (tributary of the Kheta), λ = 98°E, φ = 70.5°N, where rocks of the Lower Cretaceous (Lower and Upper Valanginian and Hauterivian) and Upper Jurassic (Lower and Upper Volga stages) were studied: 65 samples and 260 cubes were investigated;

2) on the Anabar spur in the area of the Anabar Gulf (λ = 114° E, φ = 73.5° N), also involving rocks of the Lower Cretaceous (V_{1n_1}), and the Middle and Lower Jurassic (Bathonian, Bajocian, Toarcian): 195 samples and 558 cubes;

3) the continental deposits of the Ilek suite in the Chulyma-Yenisey basin (town of Ilek, villages of Shestakovo and Kursko-Smolenskoye) (λ = 88-90.5° E, φ = 56-56.5° N), which are of Neocomian age; 147 samples were taken.

The main effort was devoted to the study of marine deposits of the Bay of Anabar, which embrace a long chronological interval and represent the most abundant collection of oriented specimens.

Because for this study it was necessary to make use of poorly cemented sands, leptochloritic sandstones, sandstones with interlayered

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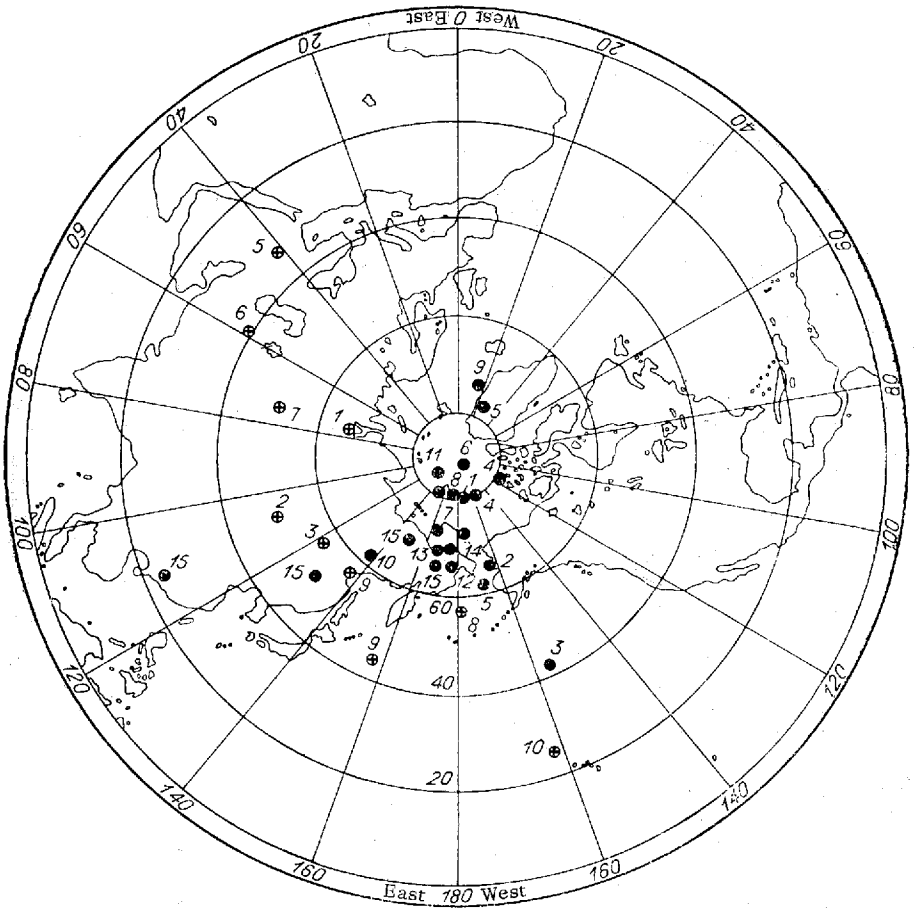


FIGURE 1. Location of the geomagnetic poles, according to paleomagnetic studies of Jurassic and Cretaceous deposits.

1 - J_1, England (17); 2, 3 - J_1, western Europe (15); 4 - J_1, North America, USA (10); 5, 6, 7 - J_1, northwest Europe (13); 8, 9 - J_1, J_2, Armenia, USSR (7); 10 - J, Taymyr Peninsula, USSR (2).

Paleomagnetic poles in Jurassic period (circles with crosses):
1 - Cr_2, North America, USA (19); 2 - Cr_2, Madagascar (18); 3 - Cr_1, Japan (16); 4 - Cr_2, England (20); 5 - Cr_2, Fergana, USSR (8); 6 - Cr_1, Czechoslovakia (9); 7, 8 - Cr_1, Central Asia, USSR (1); 9 - Cr_1, Siberia, USSR (2); 10 - Cr_1, Far East of USSR (3); 11 - Cr, Rumania (11); 12 - Cr, Fergana, USSR (5); 13 - Cr_2, California, USA; 14 - Cr_2, Korea; 15 - Cr_2, Japan.

Paleomagnetic poles in Cretaceous period (solid black circles).

gray-colored shales and siltstones, greenish-gray shales and limestones – rocks that are considered little suited for paleomagnetic studies – the primary task was to ascertain the magnetic properties of these rocks, their magnetic stability and the possibility of distinguishing their initial residual magnetization.

The studies of the natural residual magnetization (I_n) and magnetic susceptibility (χ) were made with astatic magnetometers, the ION-1 (5×10^{-8} gauss) and IMV-2 (2×10^{-6} units CGSM), whose sensitivities guaranteed sufficient accuracy and precision of the results.

The magnetic characteristics of the rocks studied are given in Table 1 for the Khatanga and Chulyma-Yenisey basins and in Figure 2 for the rocks of the Bay of Anabar.

These tabulations and diagrams show that the rocks of the Bay of Anabar, although not uniform in their magnetic properties, in general are only slightly magnetic ($I_n = 0.2 - 60 \times 10^{-6}$ gauss, $\chi = 15 - 190 \times 10^{-6}$ CGSM units), which posed great

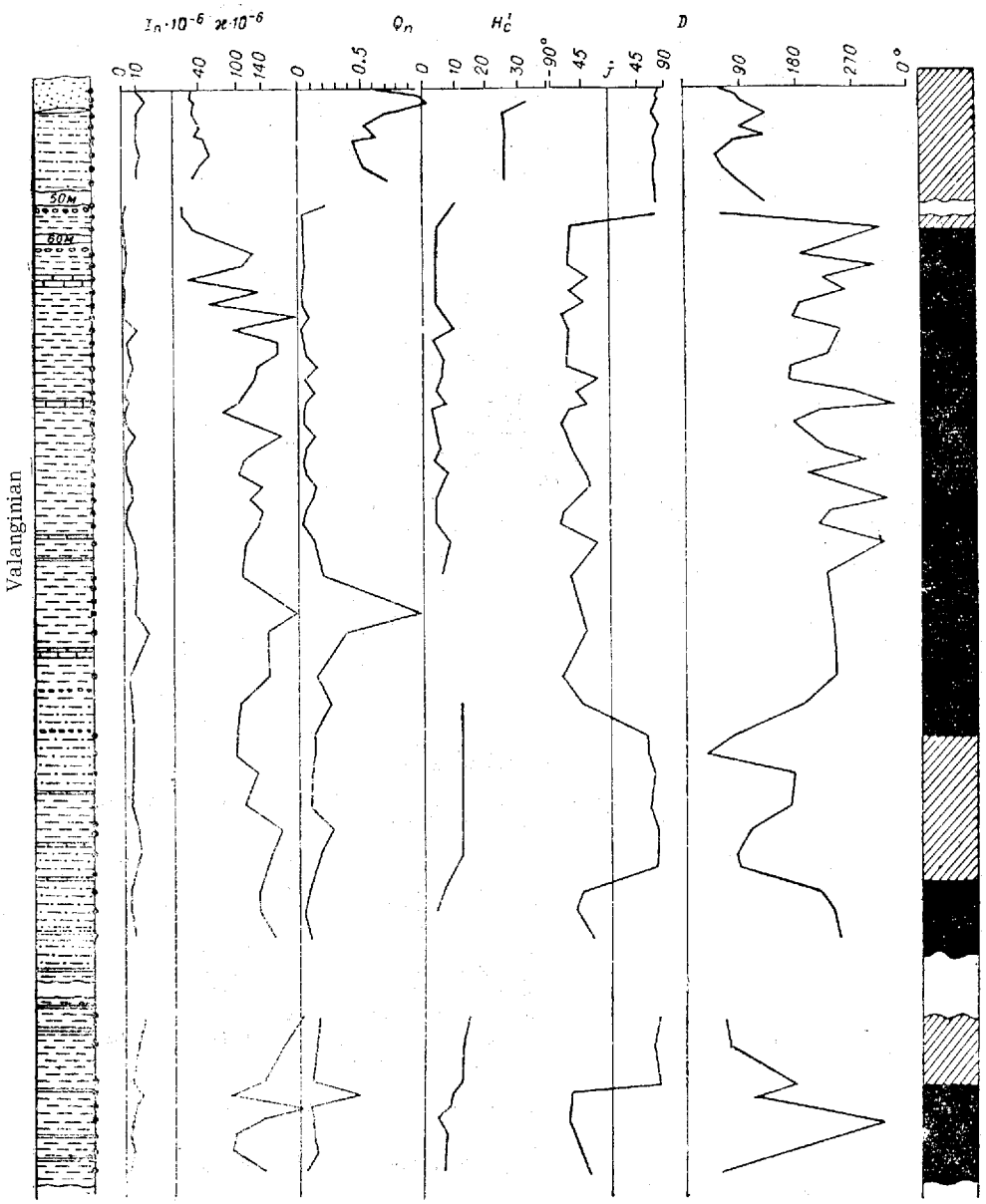


FIGURE 2. Magnetic properties of rocks from the Bay of Anabar.

1 - clayey siltstones; 2 - sandy siltstones; 3 - sandstones with shale interbeds; 4 - cross-bedded sandstones; 5 - limestones.

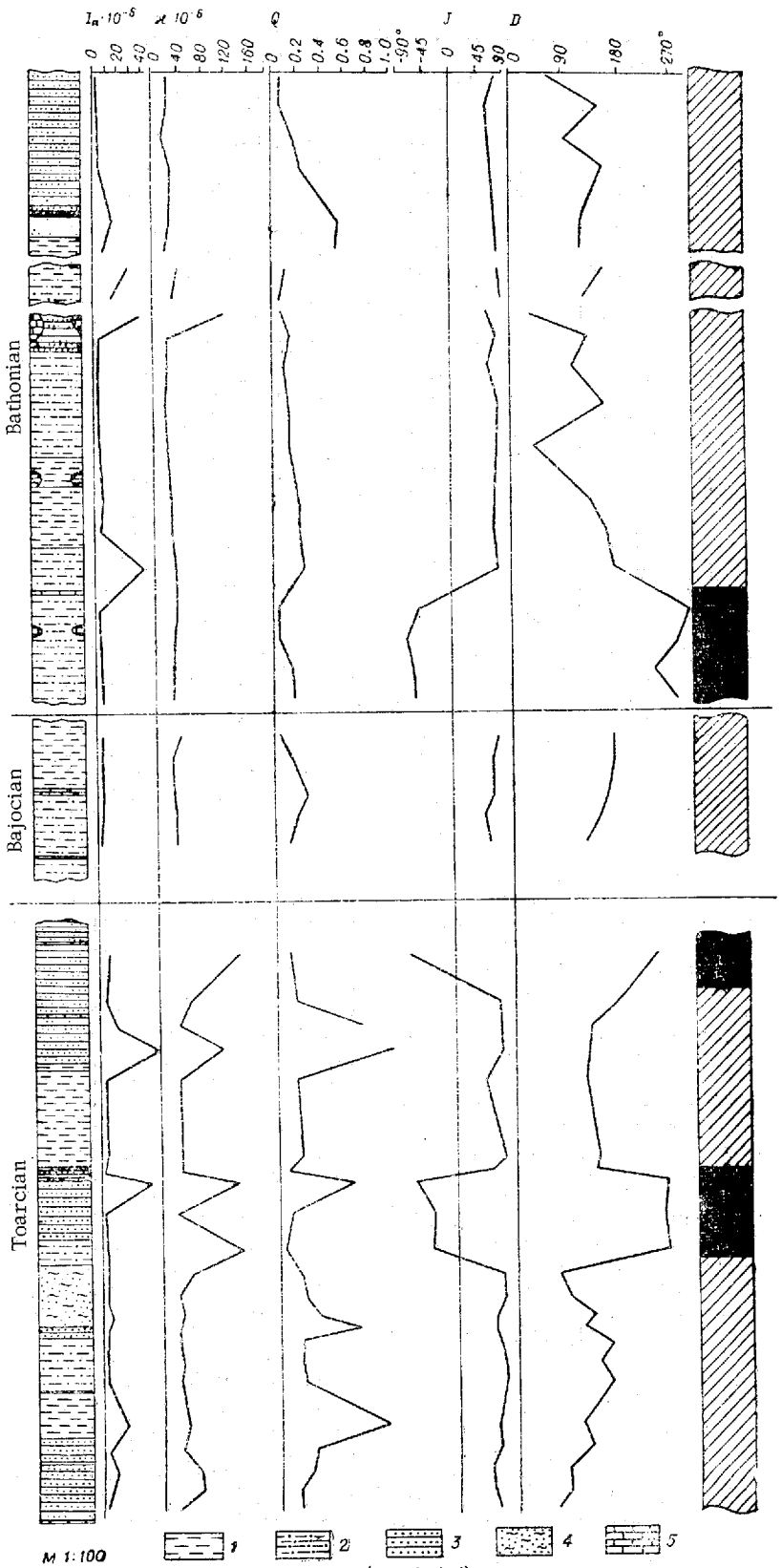


FIGURE 2 (concluded)

TABLE 1

Sample Area		Age	I_n av $\times 10^{-6}$	$\varkappa_{av} \times 10^{-6}$	Q_{av}	H'_c av	No. of Samples
Khatanga basin, Boyarka river		Lower Valanginian	20.00	180.2	0.22	8.5	6
			15.0	172	0.175	9.0	7
			2.9	151	0.039	5.5	5
			36.0	208	0.35	13.5	5
			18.0	175	0.21	11.0	7
			17.6	191	0.185	10.9	11
			16.6	137	0.25	14.1	11
			23.8	156	0.3	14.1	11
Chulyma-Yenisey basin	Kursko-Smolenskoye village	Neocomian	31.2	65.4	0.95	31.0	5
			51.2	67.3	1.5	33.5	6
			137.0	117.0	2.3	32.0	5
			33.3	101.1	0.66	27.0	3
			15.0	32.0	0.94		6
	Shestakovo village		12.5	23.0	0.53	32.0	4
			141.0	125.3	2.25		2
			14.8	20.0	1.48	39.0	8
			5.5	18.5	0.6	22.0	16
			83.0	113.0	1.45	23.0	8
	Mt. Ilek		2.52	13.4	0.38	16.2	11
			7.3	29.0	0.5	15.5	19
			18.0	24.8	0.69		2
			1.41	12.8	0.22	18.6	6
			16.48	64.0	0.52	24.0	8
			2.5	16.6	0.3	17.0	17
			31.4	123.0	0.51	21.0	4
			3.05	20.6	0.3	10.0	11
			29.0	74.0	0.76	24.0	3

difficulties in the laboratory studies.[1] The values of Q are for the most part very small, only a few tenths or hundredths of unity; only for a few isolated samples were they greater than unity. Among the rocks of the Khatanga basin the distribution of I_n and $\varkappa$ values was uniform (with the exception of one bed), and both were about one order of magnitude higher than for the rocks of the Bay of Anabar. The values of Q were also very low. The gray sandstones and siltstones of the Chulyma-Yenisey basin resemble the marine deposits in their magnetic properties. The red shales and siltstones are sharply different: I_n = = 80 - 200 x 10^{-6} gauss, Q = 1-2.5.

Study of the collapse field (H'_c) showed that it, too, is very low: from 4 to 30 oersteds, and above 30 oersteds only in occasional specimens. In each area studied, H'_c was seen to rise as the value of Q increased. This tendency is seen in all the rocks examined, but with large divergences.

The low values of I_n, Q and H'_c do not argue for the magnetic stability of the rocks studied. Of the various existing methods of estimating the stability, only a comparison of the demagnetization curves in a variable field of I_n and I_{ri} (I_{ri} being the ideal magnetization obtained from a zero magnetic state of the sample in a laboratory field in the presence of a variable magnetic field with a constantly decreasing amplitude from 700 oersteds to 0) could be applied to the rocks in question. The criterion for the suitability of these rocks was the height of the I_n curve of demagnetization as compared to the I_{ri} demagnetization curve. Comparison of the I_{ri} demagnetization curves of different samples shows that the magnetic stability of the ferromagnetic components is low, and the curves drop rapidly. But the I_n demagnetization curves of the majority of rocks lie considerably above $I_{ri} = f(H \infty)$. Figure 3 shows the I_n and I_{ri} demagnetization curves of directly and inversely magnetized samples. For the sake of comparison of the demagnetization curve with rocks characterized by natural residual magnetization having a reversed direction, it has been transformed. The computation of the curve of $I_n = f(H \infty)$ was made relative to the upper point of the curve, which continues evenly until it intersects the ordinate axis; thus it was possible to take into account the persistent

[1] Another obstacle hindering studies of most of the collection is the high latitude of sample localities and the very slight angles of inclination of the beds (< 6-16°).

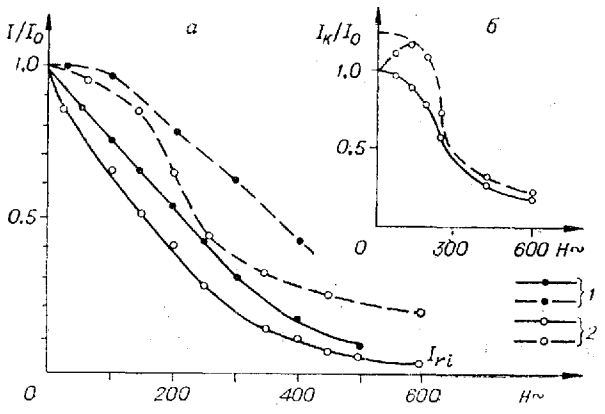


FIGURE 3

a) Comparison of I_n and I_{ri} demagnetization curves in variable fields.

b) Transformation of demagnetization curve of inversely magnetized sample.

1 - $I_n = f(\tilde{H})$ of directly magnetized sample;
2 - $I_n = f(\tilde{H})$ of inversely magnetized sample.

magnetization in the direction opposite that of I_n. The transformation of the curve of $I_n = f(H\infty)$ is shown graphically in Figure 3b. The rocks for which $I_n = f(H\infty)$ was below $I_{ri} = f(H\infty)$ were not used in further investigations.

The directions of the vectors of the natural residual magnetization of the rocks from the Bay of Anabar and the Khatanga basin were considered separately for each stratum. Examination of the stereograms indicated that the rocks were divided by the I_n vector into two groups: one group of rocks has I_n values grouped close to the direction of remagnetization (fig. 4a, c, e); the second group of rocks has an improbably large scattering of the directions of the I_n vector (fig. 4b, d, f). These groups alternate along the section. As an example, Figure 4 shows the stereograms of the distribution of I_n vectors for the Valanginian rocks of the Bay of Anabar and the Khatanga basin. The same pattern may be seen in the Jurassic rocks.

At first glance, one has the impression that the samples belonging to the first group have been completely remagnetized along the present field, and that the scattering of the vectors of the second group indicates the general unsuitability of these rocks for paleomagnetic studies. But our experience with Quaternary-Neogene clay loams having low values of Q = 0.05 and H'_c = 6-5 oersteds from the last zone of inversion, which have retained a direction of natural residual magnetization opposite that of the present geomagnetic field (which in itself testifies to the primary nature of the I_n vector and its stability) has impressed upon us the idea that the Mesozoic rocks with similar magnetic characteristics may also, if only partially, retain their primary residual magnetization (I_n^0). Detailed examination of the stereograms for the rocks of the second group suggests that the natural residual magnetization of these rocks is a vectoral sum of at least three different components: a) primary – I_n, b) secondary – chemical or persistent, arising through the prolonged effect of the geomagnetic field, and c) viscous, arising easily in the course of the laboratory investigations and just as rapidly disappearing.

Comparison of the magnetic properties of the rocks in the first and second groups shows the difference in the components of the I_n vectors. The rocks of the first group are lithologically the same as those of the second, which were sometimes taken from the very same bed but from sampling spots at different heights, and have elevated values of I_n, Q and H'_c but do not differ in χ values; this excludes any difference in the chemical processes involved in the directly and inversely magnetized samples.

The rocks were decontaminated in the laboratory, to distinguish their primary residual magnetization. According to Neyel'ya's [Neal's?] theory of persistent magnetization, the most effective form of decontamination is thermal, so that the samples were heated first of all. The decontamination was carried out in a nonmagnetic furnace at 150-200°C, in the absence of a magnetic field. Figure 5 shows the results of the thermal decontamination. The samples belonging to the first group of rocks (fig. 5a) after decontamination increased their clustering and only slightly changed their directions, whereas the samples of the second group (fig. 5b) showed a still greater dispersal of the vectors and increase in I_n values, and in some individual cases reversed their magnetic dip. Increasing the time during which the samples were heated at temperatures of 150-200°C up to 8 hours produced no significant results, except that the scattering of I_n vectors of the second group after heating diminished somewhat. But the decreased magnetic dips of I_n or even their complete reversal, and the differences in the magnetic characteristics I_n, Q and H'_c between the first and second groups of rocks indicate that the primary vectors I_n^0 of the second group are the reverse of the present geomagnetic field. It is possible that one of the secondary magnetizations in these rocks does not have a persistent, but a chemical, origin which is not effectively indicated by heating to 200°C.

The best results in the laboratory I_n decontamination of the rocks were obtained by demagnification in a variable magnetic field with maximum amplitude up to 700 oersteds. For various

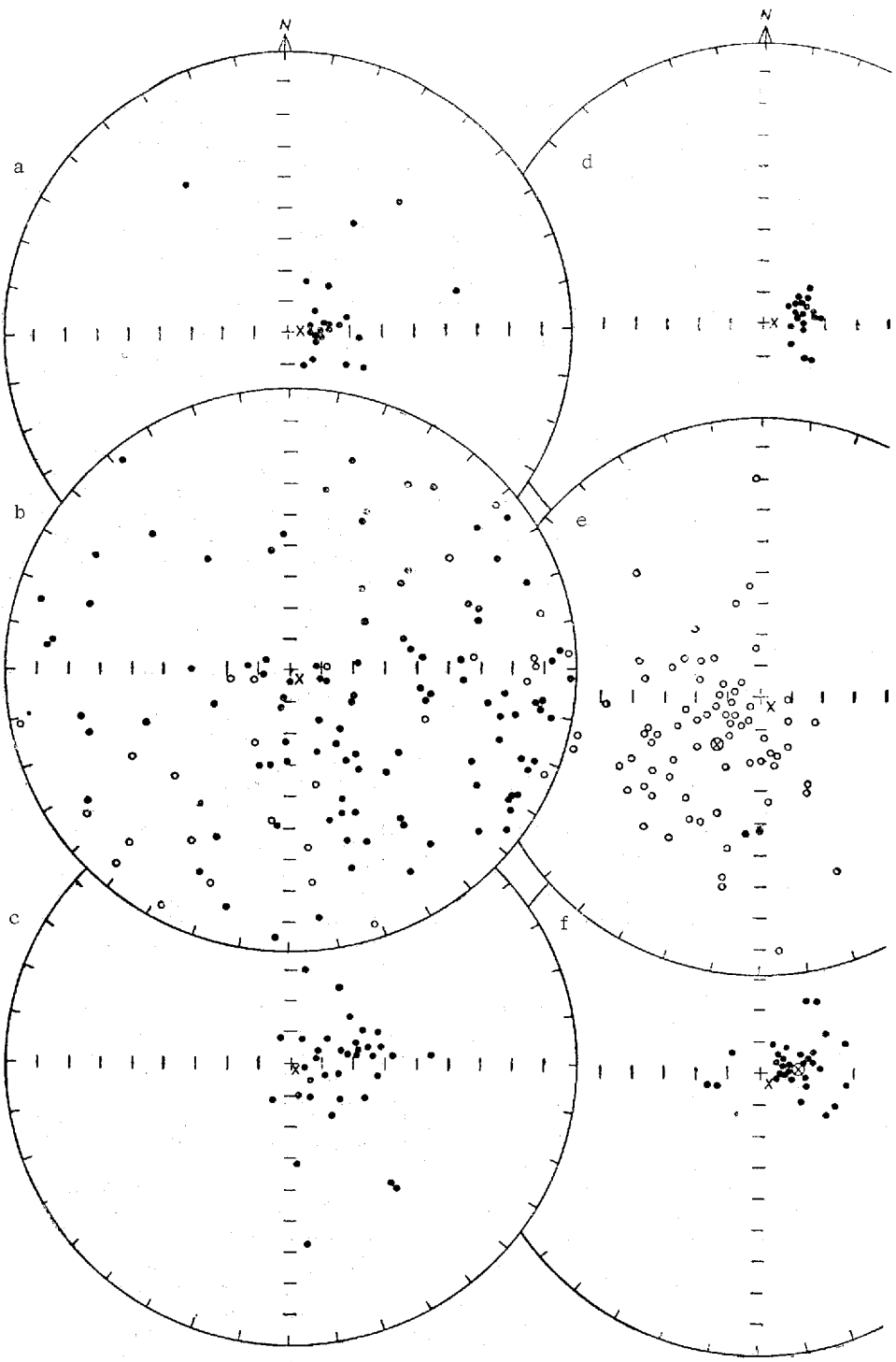


FIGURE 4. Distribution of I_n directions in Valanginian deposits.

a, b, c - before erasing (Anabar Bay); d, e, f - after erasing in variable fields (Bay of Anabar);

(Legend continued on facing page)

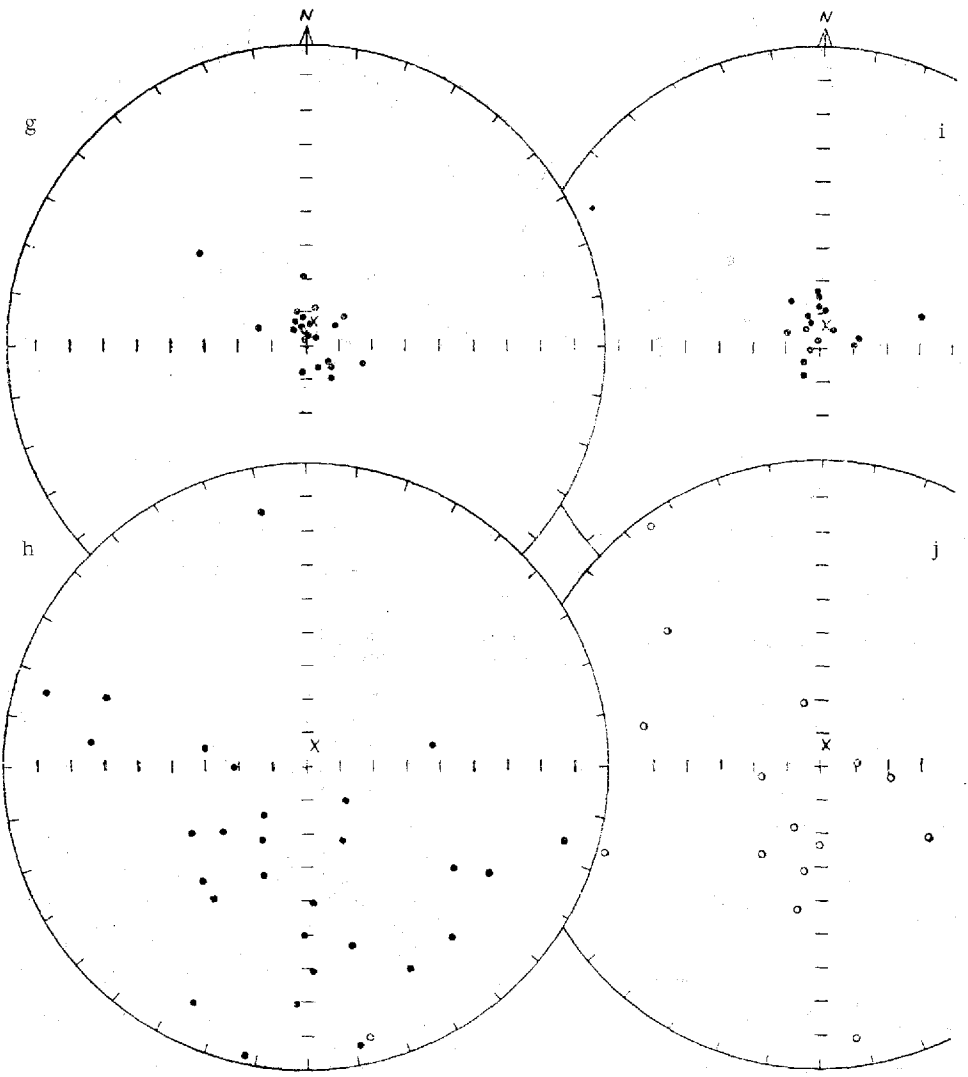


FIGURE 4 (concluded)

g, h - before erasing (Khatanga basin); i, j - after erasing in variable fields (Khatanga basin).

● - projections of I_n directions in lower hemisphere; O - same, in upper hemisphere; X - projection of direction of remagnetization; ⊗ - average direction of I_n.

samples, I_n decontamination was observed to occur in different fields from 50 to 500 oersteds. The results of decontamination of the rocks from various strata are shown in Figure 4 (d, e, f, i, j) and in Figure 5. Samples in the first group of rocks in fields up to 700 oersteds showed no significant change in the direction of I_n, although after demagnetization in a variable field the clustering of I_n increased. The average directions of I_n of these rocks are close to that of the present magnetic field in the localities from which the samples were taken, but were not quite the same. This group of rocks constitutes a zone of direct polarity. The directions of the I_n vectors of the second group of rocks after decontamination in variable magnetic fields became almost the reverse of the I_n directions in the first group. The clustering of I_n after decontamination in variable fields went from totally chaotic to about 20-50. The rocks of this second group constitute a zone of reverse polarity.

The zones of direct and reverse polarities alternate with each other along the vertical section. This occurs in both the Lower Cretaceous series and the Jurassic system. Alternation of direct

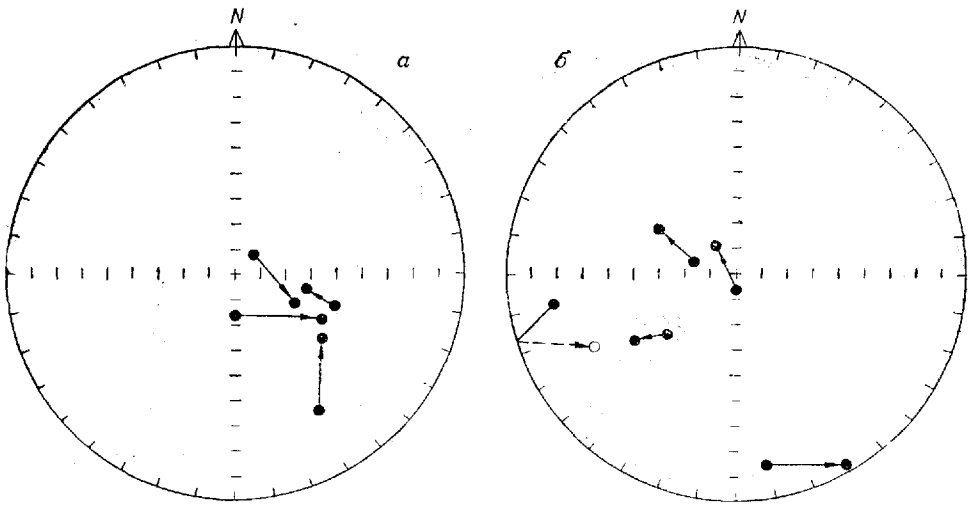


FIGURE 5. Effect of thermal decontamination on the direction of I_n.

а - decontamination of directly magnetized samples; б - decontamination of inversely magnetized samples.

→ - indicates the direction of I_n after thermal decontamination. Other symbols and designations same as in Figure 4.

and reverse magnetized zones in the Valanginian section is seen in the Bay of Anabar and the Khatanga basin sections. Of the Lower Cretaceous rocks in the Khatanga basin which according to paleontological evidence belong to the top of the Lower Valanginian stage, at the present time only one bed has been studied. In this stratum two paleomagnetic zones were found – direct and reverse; this is beautifully consistent with the paleomagnetic zonality of the Valanginian stage in the Bay of Anabar region.

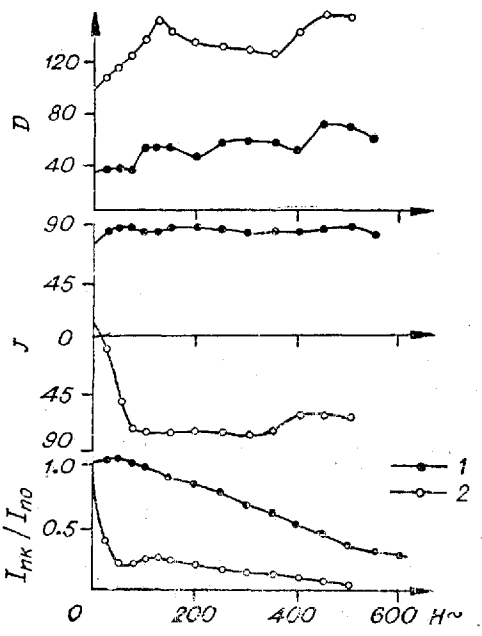


FIGURE 6. Demagnetization of I_n of samples in a variable field.

1 - directly magnetized sample;
2 - inversely magnetized sample.

Six paleomagnetic zones were found in the Lower Valanginian deposits of the Bay of Anabar. The top zone of direct magnetization has a large gap – 50 m. The samples before and after the gap are directly magnetized. Evidently this part of the section is a zone of direct magnetization. The following, negative, zone, the greatest of all in thickness, makes up almost half the section. This in turn is followed by three zones (direct – reverse – direct) which are almost equal in thickness (14, 10 and 11 m, respectively). At the bottom of this section is the last negative zone, whose lower boundary has not been located. Further studies of the Valanginian stages will be continued in geologic sections in the northern Urals, where Upper Valanginian deposits have been discovered and there is a complete and uninterrupted stratigraphic section through the Lower Valanginian and the Berriasian.

The alternation of paleomagnetic zones was first discovered in the Valanginian stage. It seems likely that the duration of a single paleomagnetic zone was about 500,000 years, as with the paleomagnetic zones of the Pliocene series (6). The existence of paleomagnetic zones in two different areas (Khatanga basin and the Bay of Anabar) testifies to inversions of the geomagnetic field and appears to be an additional criterion for the stratigraphic correlation of the deposits in the different regions.

TABLE 2

Zones	J^0_{av}	D^0_{av}	λ^0 E long	Φ^0 N lat	α^0	θ^0_1	θ^0_2	φ^0_m
1	77.5	85.5	174	62	2.5	4.8	6.5	66
3	79.0	87.0	170	63.5	3.0	5.65	5.3	68.5
5	77.45	78.0	177	64	6.5	11.8	11.1	66.5

The rocks of the Ilek suite, which geologists have assigned to the Neocomian, were comparable with the paleomagnetic section through the Valanginian in the Bay of Anabar region. But no zones of inverse polarity were found in the deposits of the Ilek suite in the Chulyma-Yenisey basin, indicating the absence of Lower Valanginian sediments in this sequence.

The Jurassic deposits were also studied in the areas of the Khatanga basin and the Bay of Anabar. But these are of different ages and cannot be correlated: Upper Jurassic rocks studied in the Khatanga basin and Middle and Lower Jurassic in the Bay of Anabar area.

No alternation of paleomagnetic zones was found in the deposits of the Volga stage, and all rocks were magnetized positively. But in the sections through the Middle and Lower Jurassic on the shores of the Bay of Anabar, an alternation of paleomagnetic zones was discovered again. Two paleomagnetic zones were found in the Bathonian stage; of these the upper, or positive, zone occupies the greater part of the section (70 m), and only the base of this section is represented by a small negative zone (10 m). As much of the upper part of the Bajocian stage as was studied was paleomagnetically positive. In the Upper and Middle Toarcian there were four paleomagnetic zones, the reverse or negative zones being quite thin (4 m and 8 m). The upper negative zone probably continues upward along the section. The greater part of the deposits is represented by directly magnetized rocks (positive zones 15 m and 20.5 m thick) (fig. 2).

The locations of the ancient geomagnetic poles were calculated from the average I_n vectors after decontamination in variable fields for each directly magnetized zone of the Valanginian stage in the Bay of Anabar region (table 2). The $\vec{I}_n$ vectors of the inversely magnetized rocks were not used in calculating the locations of the poles, since the I_n values of these rocks could not be regarded as primary because of their incomplete decontamination.

The average locations of the poles during the Early Valanginian age were: $\lambda_{av} = 174^\circ$ E, $\varphi_{av} = 63.2^\circ$ N. The determinations of the paleolatitudes of the sample localities ($\lambda = 114^\circ$ E, $\varphi = 73.5^\circ$ N) for the Valanginian age by the paleomagnetic method produced results that agree nicely with those yielded by the paleontological data: $\varphi_m = 67^\circ$ N.

The main results of our investigations may be summed up as follows:

1. The range of rock types that may be considered suitable for paleomagnetic studies has been considerably expanded. After laboratory decontamination, one can confidently make use of such sedimentary rocks as sandstones, limestones, gray-colored clays and shales, siltstones and poorly cemented sands – rocks possessing high magnetic viscosity, low values of Q (tenths and hundredths of unity) and low values of the collapse field (down to 4 oersteds).

2. One criterion for the paleomagnetic suitability of a rock is a high I_n demagnetization curve as compared to the I_{ri} curve.

3. The results of our research have added to the scanty existing data on the history of the geomagnetic field in the Jurassic and Cretaceous periods.

4. Alternations of zones of direct and reverse magnetization were found in the Valanginian, Bathonian and Toarcian stages; these alternating zones may be used for the chronological correlation of sedimentary sequences.

In considering the results of these investigations, we have assumed that the primary natural remanent magnetization of the rocks studied is oriented in nature. But the present authors will carry out further investigations of the magnetic properties of rocks for a final determination of the nature of remanent magnetization.

In conclusion, the writers wish to express their sincere gratitude to V. N. Saks, corresponding member of the USSR Academy of Sciences, for his valuable advice, and to Z. L. Kulesheva who took part in the laboratory treatment and investigation of the rock samples.

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